STEP 1 — Create a new notebook/script

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2403A52379_WordEmbeddings_LAB-09_NLP

Step-2 Import Required Libraries

```
# @title Step-2 Import Required Libraries
# Gensim is used to load and work with pre-trained Word2Vec embeddings
!pip install gensim
from gensim.models import KeyedVectors

# NumPy is used for numerical operations on vectors
import numpy as np

# sklearn is used for dimensionality reduction (PCA / t-SNE)
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE

# matplotlib is used for visualization of word vectors
import matplotlib.pyplot as plt
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1-
```

STEP 3 — Load Pre-trained Embeddings

```
# @title STEP 3 — Load Pre-trained Embeddings
# Path to downloaded Google News Word2Vec model
# Rename the file to ensure gensim recognizes the .gz extension for decompression
!mv GoogleNews-vectors-negative300.bin.gz.1 GoogleNews-vectors-negative300.bin.gz
model_path = "GoogleNews-vectors-negative300.bin.gz"

# Load the Word2Vec model (binary=True is required)
word_vectors = KeyedVectors.load_word2vec_format(model_path, binary=True, encoding='latin-1')

# Print vocabulary size
print("Vocabulary size:", len(word_vectors.key_to_index))

# Display example word vectors
print("\nVector for word 'king':")
print(word_vectors["king"][:10])  # printing first 10 values only
```

```
Vocabulary size: 3000000

Vector for word 'king':
[ 0.12597656  0.02978516  0.00860596  0.13964844 -0.02563477 -0.03613281
  0.11181641 -0.19824219  0.05126953  0.36328125]
```

```
!wget https://github.com/mmihaltz/word2vec-GoogleNews-vectors/raw/master/GoogleNews-vectors-negative300
```

```
--2026-02-10 04:24:15--  https://github.com/mmihaltz/word2vec-GoogleNews-vectors/raw/master/GoogleNews-v
Resolving github.com (github.com)... 140.82.112.3
Connecting to github.com (github.com)|140.82.112.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://media.githubusercontent.com/media/mmihaltz/word2vec-GoogleNews-vectors/master/GoogleNe
--2026-02-10 04:24:15--  https://media.githubusercontent.com/media/mmihaltz/word2vec-GoogleNews-vectors/
Resolving media.githubusercontent.com (media.githubusercontent.com)... 185.199.111.133, 185.199.109.133,
Connecting to media.githubusercontent.com (media.githubusercontent.com)|185.199.111.133|:443... connecte
HTTP request sent, awaiting response... 200 OK
```

```
Length: 1647046227 (1.5G) [application/octet-stream]
Saving to: ‘GoogleNews-vectors-negative300.bin.gz.1’

GoogleNews-vectors- 100%[===================>]   1.53G  12.0MB/s    in 79s

2026-02-10 04:25:34 (20.0 MB/s) - ‘GoogleNews-vectors-negative300.bin.gz.1’ saved [1647046227/1647046227
```

STEP 4 — Explore Word Similarity

```
# @title STEP 4 — Explore Word Similarity
# List of word pairs to compute similarity
word_pairs = [
    ("doctor", "nurse"),
    ("cat", "dog"),
    ("car", "bus"),
    ("king", "queen"),
    ("apple", "orange"),
    ("india", "china"),
    ("teacher", "student"),
    ("computer", "keyboard"),
    ("football", "cricket"),
    ("movie", "film")
]

# Compute cosine similarity for each pair
for w1, w2 in word_pairs:
    similarity = word_vectors.similarity(w1, w2)
    print(f"Similarity between {w1} and {w2}: {similarity:.3f}")
```

```
Similarity between doctor and nurse: 0.632
Similarity between cat and dog: 0.761
Similarity between car and bus: 0.469
Similarity between king and queen: 0.651
Similarity between apple and orange: 0.392
Similarity between india and china: 0.353
Similarity between teacher and student: 0.630
Similarity between computer and keyboard: 0.396
Similarity between football and cricket: 0.460
Similarity between movie and film: 0.868
```

STEP 5 — Nearest Neighbor Exploration

```
# @title STEP 5 — Nearest Neighbor Exploration
from gensim.models import KeyedVectors

model_path = "GoogleNews-vectors-negative300.bin.gz"
word_vectors = KeyedVectors.load_word2vec_format(model_path, binary=True, encoding='latin-1')

# Words to explore nearest neighbors
query_words = ["king", "university", "doctor", "computer", "india"]

for word in query_words:
    print(f"\nTop similar words to '{word}':")
    for similar_word, score in word_vectors.most_similar(word, topn=5):
        print(f"  {similar_word} : {score:.3f}")
```

```

Top similar words to 'king':
  kings : 0.714
  queen : 0.651
  monarch : 0.641
  crown_prince : 0.620
  prince : 0.616

Top similar words to 'university':
  universities : 0.700
  faculty : 0.678
  unversity : 0.676
  undergraduate : 0.659
```

```
  univeristy : 0.659

Top similar words to 'doctor':
  physician : 0.781
  doctors : 0.748
  gynecologist : 0.695
  surgeon : 0.679
  dentist : 0.679

Top similar words to 'computer':
  computers : 0.798
  laptop : 0.664
  laptop_computer : 0.655
  Computer : 0.647
  com_puter : 0.608

Top similar words to 'india':
  indian : 0.697
  usa : 0.684
  pakistan : 0.682
  chennai : 0.668
  america : 0.659
```

STEP 6 — Word Analogy Tasks

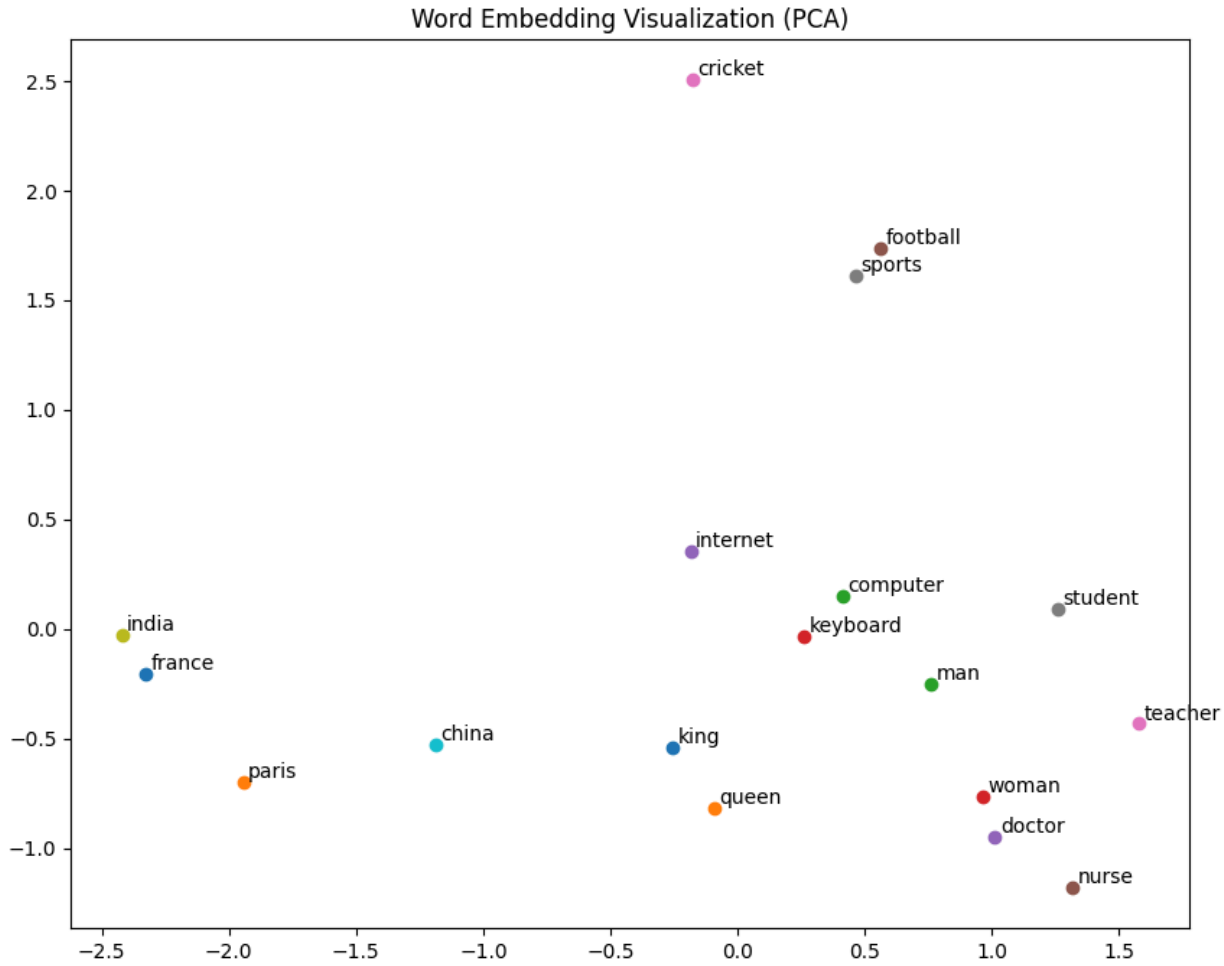
```
# @title STEP 6 — Word Analogy Tasks
# Analogy queries using vector arithmetic
analogies = [
    ("king", "man", "woman"),
    ("paris", "france", "india"),
    ("teacher", "school", "hospital")
]

for a, b, c in analogies:
    result = word_vectors.most_similar(
        positive=[a, c],
        negative=[b],
        topn=1
    )
    print(f"{a} - {b} + {c} = {result[0][0]}")
```

```
king - man + woman = queen
paris - france + india = chennai
teacher - school + hospital = Hospital
```

STEP 7 — Visualization

```
# @title STEP 7 — Visualization
import numpy as np
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Select words for visualization
words = [
    "king", "queen", "man", "woman",
    "doctor", "nurse", "teacher", "student",
    "india", "china", "france", "paris",
    "computer", "keyboard", "internet",
    "football", "cricket", "sports"
]

# Extract vectors
X = np.array([word_vectors[word] for word in words])

# Reduce dimensions using PCA
pca = PCA(n_components=2)
X_reduced = pca.fit_transform(X);

# Plot
plt.figure(figsize=(10, 8))
```

```
for i, word in enumerate(words):
    plt.scatter(X_reduced[i, 0], X_reduced[i, 1])
    plt.text(X_reduced[i, 0] + 0.02, X_reduced[i, 1] + 0.02, word)

plt.title("Word Embedding Visualization (PCA)")
plt.show()
```



Close

Double-click (or enter) to edit

STEP 8 — Reflection and Interpretation

Word embeddings learn semantic relationships by analyzing word co-occurrence patterns in large text corpora. Similar words appear close together in vector space, such as “doctor” and “nurse”. The model captures analogical relationships using vector arithmetic, demonstrating structured semantic knowledge. Geographic and gender relationships are particularly strong. However, embeddings sometimes fail for rare or ambiguous words. Biases present in training data can also appear in embeddings. Context is not dynamic in Word2Vec, so a word has only one meaning. This causes problems for polysemous words. Despite limitations, embeddings are powerful representations of language. They are widely used in NLP tasks like translation and search.

STEP 9 — Lab Report Structure

Objective:

To explore semantic relationships using pre-trained Word2Vec embeddings.

Model / Dataset

Google News Word2Vec (300D vectors, ~3 million words)

Experiments:

Word similarity

Nearest neighbors

Analogy tasks

Visualization

Observations:

Similar words have high cosine similarity

Analogies work via vector arithmetic

Semantic clusters emerge clearly

Conclusion:

Word embeddings effectively capture semantic meaning but lack contextual awareness.